

Digital Evidence & Forensic Incident Dossier

REPORT REFERENCE:	RPT-█20260908-█18AEAE	GENERATED (UTC):	2026-█09-█08 10:█16:█42 UTC
EVIDENCE IDENTIFIER:	EVD-█5E58C282C0	ASSIGNED CASE:	a491dcb7-█a74b-█4cb0-█b7a4-█4effd6bc9b7c
INVESTIGATOR:	SOC Lead Analyst	CLASSIFICATION:	CONFIDENTIAL // EVIDENTIARY DISCLOSURE
EMAIL SHA-256:	5e58c282c0ac76456bf139f43214█0e0dfc5338ca13502c9bea990d35█aa825c3b	HASH ALGORITHM:	FIPS PUB 180-4 SHA-256 (Preserved Literal)

1. Executive Forensic Summary

DETERMINISTIC THREAT SCORE	ASSESSED SEVERITY	ML PHISHING PROBABILITY	ML CLASSIFICATION
38/100	SUSPICIOUS	65.5%	PHISHING

Email risk assessed as Suspicious (38/100) driven by 3 forensic signal(s): Possible PayPal lookalike domain (+18), Suspicious URL detected (+8), Credential harvesting or urgent security language (+12).

2. Email Headers & Sender Alignment

From:	security-█alert@█paypal-█update.█com
To:	victim@█company.█com
Subject:	Urgent Security Alert:█ Account Suspicious Access
Date:	Tue,█ 08 Sep 2026 10:█30:█00 +█0000
Reply-To:	None
Return-Path:	None
Message-ID:	&█lt;█evd-█test-█12345@█paypal-█update.█com&█gt;█

From Domain:	paypal-█update.█com	Reply-To Domain:	N/█A (Aligned)
Return-Path Domain:	N/█A (Aligned)	Alignment Verdict:	Sender domains aligned

3. Email Authentication Findings (SPF / DKIM / DMARC)

PROTOCOL	OBSERVED VERDICT	TECHNICAL DETAILS & EVIDENCE
SPF (Sender Policy)	NONE	No Authentication-█Results or Received-█SPF header present.█
DKIM (Cryptographic)	NONE	No Authentication-█Results or DKIM-█Signature header present.█
DMARC (Domain Policy)	NONE	No Authentication-█Results header present.█

Notice: Observed authentication result from supplied headers (unverified by local mail server)

4. Deterministic Threat Score Breakdown

CATEGORY	IMPACT	FORENSIC FINDING & JUSTIFICATION
General	+18 pts	N/A

General	+8 pts	N/A
General	+12 pts	N/A

5. NLP Content Phishing Classification

Model Name:	TF-IDF + Logistic Regression	Classification:	PHISHING
Phishing Probability:	65.5%	Model Confidence:	medium
Top Text Features:	httpurl, account, immediately, security, urgent	Signal Role:	Auxiliary ML signal (Non-overriding)

6. AI Analyst Assistant Synthesis

Likely Attack Type:	Credential Harvesting Phishing	Likely Objective:	Theft of corporate directory or single-sign-on credentials via spoofed interfaces
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Executive Synthesis: The analyzed email presents characteristics of Credential Harvesting Phishing with a deterministic threat score of 38/100 (SUSPICIOUS). Key driving indicators include 3 forensic flag(s). Adversary objective is assessed as: Theft of corporate directory or single-sign-on credentials via spoofed interfaces.

Key Forensic Evidence Identified by LLM:

- Global threat score evaluated at 38/100 (SUSPICIOUS severity).
- Forensic indicator: Possible PayPal lookalike domain
- Forensic indicator: Suspicious URL detected
- Forensic indicator: Credential harvesting or urgent security language
- Brand lookalike detected: 'paypal-update.com' targeting 'PayPal'.

7. Extracted Indicators of Compromise (IOCs)

Observed URLs & Suspicion Scoring:

EXTRACTED URL / TARGET HOST	SCORE	FORENSIC OBSERVATIONS & FLAGS
http://paypal-update.com/login	30/100	Potential brand impersonation: similar to paypal.com (90% match), Suspicious security/authentication keywords found: login, update

Observed Attachments & Cryptographic Checksums:

No file attachments detected in this message.

8. Network Transmission Path & Topology

Earliest Observable Sending IP:	None identified	Reverse DNS / Host:	Unknown
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No sequential relay hops reconstructed from Received: headers.

Notice: Labeled strictly as observed infrastructure route. Does not represent physical attacker travel path.

9. Infrastructure & Domain Intelligence

No public IP intelligence enriched for this email.

10. Brand Impersonation & Lookalike Domains

SUSPECTED BRAND	SPOOFED DOMAIN	SIMILARITY	TECHNIQUES & ATTACK PROFILE
PayPal	paypal-█update.█com	90.0%	hyphenation,█ brand_█keyword,█ added_█affix

11. Prescriptive Recommendations & Containment Actions

- [] 1. Tag message header with [SUSPICIOUS] warning banner for internal recipient.
- [] 2. Monitor user account for anomalous authentication or link traversal events.

12. Evidentiary Limitations & Chain-of-Custody Certification

Forensic Chain-of-Custody Certification:
This document is generated by MailTrace AI based on literal RFC-822 evidence headers. Public IP geolocation represents coarse regional estimates based on RIR allocations and does not establish street-level physical locations or the personal identity of the sender. All cryptographic hash sums (SHA-256) were computed over unaltered message byte streams and preserve complete digital non-repudiation chain of custody.

EVIDENTIARY DIGITAL SEAL:	SHA256:5e58c282c0ac76456bf139f43214 █0e0dfc5338ca13502c9bea990d35█aa825 c3b	VERIFICATION STATUS:	CRYPTOGRAPHICALLY VERIFIED
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